

Kaylena Pham

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EDUCATION

University of Southern California
Bachelor of Science in Geological Sciences
Major GPA: 3.95/4.00

August 2023 – Expected May, 2027
Los Angeles, CA

RESEARCH EXPERIENCE

USC Department of Earth Sciences – Undergraduate Researcher

October 2023—

Atmospheric Composition and Earth Data Science Group

Advisor: Sam Silva

- Studied physiological resilience of saguaro cacti to elevated ozone levels by analyzing stomatal conductance and resistance data, and parameterized bulk surface resistance through sensitivity analysis across land-usages and seasons
- Processed and visualized time-series atmospheric ozone and plant physiological data using Python and Pandas, applying data analysis techniques to identify trends and anomalies in gas exchange behavior

Significant Opportunities in Atmospheric Research and Science (SOARS) Protégé (Research Intern)

NSF National Center for Atmospheric Research

May – July 2026

Advisors: Shima Shams, Kathryn Payne, William Nicewonger

- Characterize and correct low-cost 3D-PAWS Plantower PMSA003I PM2.5 sensors against Teledyne reference-grade instruments across three field sites (Athens, GA; Boulder, CO; Lee District Park, VA), improving low-cost sensor data reliability for community air quality monitoring
- Synthesized calibration and QA/QC literature to inform correction methodology and identify best practices for low-cost PM2.5 sensor networks
- Presented research at Annual UCAR Student Intern Poster and Oral Presentation, connecting sensor calibration work to broader environmental justice goals in air quality data access

Student Airborne Research Program (SARP) Research Intern

June – August 2025

National Aeronautics and Space Program (NASA)

[“Spooky Swamps: How Methane Emission Rates and Their Spatial Variability Differ Between the Great Dismal Swamp and the Alligator River”](#)

Advisor: Stacey Hughes

- Investigated spatial variability of methane emission rates between the Great Dismal Swamp and Alligator River wetland systems using airborne remote sensing data
- Integrated MODIS NDVI satellite imagery with in-situ methane measurements from NASA SARP flight campaigns and conducted comparative analyses between multiple field sites, incorporating variables such as soil moisture and canopy cover
- Delivered oral presentation synthesizing flux measurements and spatial analysis of wetland methane sources

CONFERENCES AND PRESENTATIONS

- 2027. (Submitted)** Characterizing PM2.5 Data in Low-Cost Air Quality Sensors. American Meteorological Society 2027 Annual Meeting. Denver, Colorado. January. [with Shima Shams, William Nicewonger, and Kathryn Payne]

- **2026. (Submitted)** The Path of Most Resistance: A Sensitivity Analysis of Bulk Surface Resistance to Land-Use Type and Seasonal Variability in Ozone Dry Deposition. American Geophysical Union 2026 Annual Meeting. San Francisco, California. January. [with Sam Silva and Brian Schlaff]
- **2025. (Oral)** Spooky Swamps: How Methane Emission Rates and Their Spatial Variability Differ Between the Great Dismal Swamp and the Alligator River. American Geophysical Union 2025 Annual Meeting. New Orleans, Louisiana. December. [with Stacey Hughes and Katherine Paredero]
- **2025. (Poster)** Investigating the Impact of Ozone Dry Deposition on Saguaro Cacti (*Carnegiea Gigantea*) in Saguaro National Park, Arizona. American Meteorological Society 2025 Student Conference. New Orleans, Louisiana. December. [with Sam Silva and Brian Schlaff]

LEADERSHIP ACTIVITIES

Queers in Science, Engineering, and Technology (QuEST)

August 2024—

President (August 2025 – Present)

Conference Chair (August 2024 – May 2025)

- Lead a 100+ member USC oSTEM chapter as President, overseeing professional development programming, corporate outreach, and community-building initiatives for LGBTQ+ students in STEM
- Served as Conference Chair (2024–2025), organizing funding and logistics for students to attend the annual oSTEM conference to connect members with industry and academic opportunities

Out for Undergrad – Engineering and Sciences (O4U E&S)

January 2025 – May 2025

Conference Campus Ambassador

- Recruited and promoted O4U's national conference to LGBTQ+ students in engineering and science at USC through campus outreach and tabling events
- Mentored two students through the conference application process, providing guidance to support their successful admission to the conference

Student Sustainability Committee (SSC)

October 2023 – May 2024

Project Co-Lead

- Co-led a 10-person team developing a database to track food purchased, consumed, and discarded across campus dining halls, in partnership with university administration to reduce campus food waste
- Designed a staff-facing data input system that automatically calculates food waste relative to purchases, solving a key data collection challenge requiring ongoing coordination with dining hall staff

AWARDS

- USC LGBTQ+ Lambda Scholar (2024 – 2027)
- Women in Mining California Chapter Scholar (2024)
- Presidential Scholarship (2023 – 2027)
- Dean's List (Fall 2023, Fall 2025, Spring 2026)

SKILLS

- Proficient in Python for data and statistical analysis
 - **Data Analysis & Manipulation:** Numpy, Pandas, XArray, Scikit-learn
 - **Statistical:** SciPy, Statsmodels
 - **Visualization:** Matplotlib, Seaborn, Cartopy
 - **General Computing:** Jupyter/Jupyter Hub, Pangeo, GitHub